A PROOF OF KARTTUNEN'S CONJECTURE A401714$(n) \le$ A401715(n), WITH A COMPLETE FORMAL VERIFICATION IN LEAN 4

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Abstract. For a positive integer n, let $\sigma^*(n)$ denote the sum of the unitary divisors of n, let $\sigma_\square(n)$ denote the sum of its squarefree divisors, let $\mathrm{rad}(n)$ be the product of the distinct primes dividing n, and let $q(n)$ be the smallest prime not dividing n. We prove that

$$\frac{n}{\mathrm{rad}(n)} < q(n) \quad \Longrightarrow \quad \sigma^*(n) + \sigma_\square(n) \ge 2n.$$

This settles a conjecture recorded by A. Karttunen in the OEIS on November 6, 2026, namely A401714$(n) \le$ A401715(n) for all $n \ge 1$, equivalently that A388986 is a subsequence of A393869. The bound is sharp, and under the hypothesis equality holds exactly for $n \in \{1, 4, 24\}$. The proof is elementary. For $q(n) \le 73$ it reduces to 691 explicit finite verifications. For $q(n) > 73$ it rests on the inequality $\prod_{p<x}(1 + 1/p) \ge \frac{1}{2}H_{x-1}$ and on a rearrangement argument. The main inequality has been formalized and machine-checked in Lean 4 with Mathlib; the classification of the equality cases is established separately, by the written argument and an exact finite computation.

1. Introduction

Throughout, p always denotes a prime, n a positive integer, and $v_p(n)$ the exponent of p in n. We use the following arithmetic functions:

- $\sigma^*(n) = \sum_{d\mid n,\ \gcd(d,n/d)=1} d$, the sum of the unitary divisors of n (A034448);
- $\sigma_\square(n) = \sum_{d\mid n,\ d \text{ squarefree}} d$, the sum of the squarefree divisors of n (A048250);
- $\mathrm{rad}(n) = \prod_{p\mid n} p$, and $A(n) = n/\mathrm{rad}(n)$ (A003557);
- $q(n)$, the smallest prime not dividing n (A053669).

In the On-Line Encyclopedia of Integer Sequences, A401714 is the characteristic function of the condition $A(n) < q(n)$, and A401715 is the characteristic function of the condition $\sigma^*(n)+\sigma_\square(n) \ge 2n$. Karttunen conjectured that A401714$(n) \le$ A401715(n) for all $n \ge 1$ [1, 2]. Since both sequences take only the values 0 and 1, this is equivalent to the following statement, which we prove.

Theorem 1.1. *For every integer $n \ge 1$,*

$$A(n) < q(n) \quad \Longrightarrow \quad \sigma^*(n) + \sigma_\square(n) \ge 2n.$$

Equivalently, the set A393981 of k with $A(k) < q(k)$ is contained in the set A389215 of k with $\sigma^*(k) + \sigma_\square(k) \ge 2k$. Since $A(n)$ divides n while $q(n)$ does not, $A(n) \ne q(n)$ for every n, so the complement of the condition $A(n) < q(n)$ is $A(n) > q(n)$. Passing to complements, A388986 is a subsequence of A393869; this is the form in which the conjecture is stated on A388986 [3].

The inequality cannot be improved. For $n = 4$ we have $A(4) = 2 < 3 = q(4)$ and $\sigma^*(4) + \sigma_\square(4) = 5 + 3 = 8 = 2 \cdot 4$. Corollary 6.1 shows that, under the hypothesis of Theorem 1.1, equality holds exactly for $n \in \{1, 4, 24\}$. The hypothesis cannot simply be dropped: $n = 240$ satisfies $\sigma^*(240) + \sigma_\square(240) = 408 + 72 = 480 = 2 \cdot 240$ but has $A(240) = 8 > 7 = q(240)$, and there are n (for instance $n = 8$) with $\sigma^*(n) + \sigma_\square(n) < 2n$.

Structure of the proof. After the preliminaries of Section 2, Section 3 reduces the problem to numbers of the form $n_0 = a \cdot \prod_{p<q} p$ with $1 \le a < q$. Section 4 treats $q \le 73$ by direct

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verification. Section 5 treats $q > 73$: there the single term $\sigma^*(n)$ is already at least $2n$, and this follows from an elementary lower bound for $\prod_{p<x}(1+1/p)$. Section 7 describes the formal verification.

2. Preliminaries

For $n \geq 1$ write $n = \prod_{p\in S} p^{e_p}$, where $S = S(n)$ is the set of primes dividing n and $e_p = v_p(n) \geq 1$. Then

$$\mathrm{rad}(n) = \prod_{p\in S} p, \qquad A(n) = \prod_{p\in S} p^{e_p-1}.$$

Lemma 2.1. *For every $n \geq 1$,*

$$\sigma^*(n) = \prod_{p\in S}(1+p^{e_p}), \qquad \sigma_\square(n) = \prod_{p\in S}(1+p).$$

Proof. Let $d \mid n$. If $\gcd(d, n/d) = 1$, then for each $p \in S$ the prime p cannot divide both d and n/d, so $v_p(d) \in \{0, e_p\}$. Hence the unitary divisors of n are exactly the numbers $d_U = \prod_{p\in U} p^{e_p}$ with $U \subseteq S$. Conversely, each d_U is a unitary divisor, since $n/d_U = d_{S\setminus U}$ and d_U, $d_{S\setminus U}$ have no prime factor in common. Distinct U give distinct d_U, because the set of prime factors of d_U is U. Therefore $\sigma^*(n) = \sum_{U\subseteq S}\prod_{p\in U} p^{e_p} = \prod_{p\in S}(1+p^{e_p})$. In the same way, the squarefree divisors of n are exactly the numbers $\prod_{p\in U} p$ with $U \subseteq S$, all distinct, which gives the formula for $\sigma_\square(n)$. □

Remark 2.2. The formal proof of the main inequality uses only the lower bounds $\sigma^*(n) \geq \prod_{p\in S}(1+p^{e_p})$ and $\sigma_\square(n) \geq \prod_{p\in S}(1+p)$, which follow from the injectivity of $U \mapsto d_U$ alone. Wherever the identities of Lemma 2.1 are used below in the proof of Theorem 1.1 (in (1) and in Lemmas 3.1 and 5.1), only the corresponding lower bounds are needed. The identities themselves are needed for the classification of the equality cases (Corollary 6.1), which is not part of the formalization.

Dividing by n, Lemma 2.1 gives

$$F(n) := \frac{\sigma^*(n)+\sigma_\square(n)}{n} = \prod_{p\in S}\Big(1+\frac{1}{p^{e_p}}\Big) + \frac{1}{A(n)}\prod_{p\in S}\Big(1+\frac{1}{p}\Big), \tag{1}$$

and Theorem 1.1 asserts that $F(n) \geq 2$ whenever $A(n) < q(n)$.

Lemma 2.3. *Let $n \geq 1$ with $A(n) < q(n)$. Put $q = q(n)$ and $T = \{p \in S : e_p \geq 2\}$. Then:*

(i) *every prime $p < q$ divides n, and $q \nmid n$;*
(ii) *$\prod_{p\in T} p \leq A(n) < q$; in particular every $p \in T$ satisfies $p < q$;*
(iii) *every prime $p > q$ dividing n satisfies $e_p = 1$.*

Proof. (i) holds by the definition of q. For (ii), every factor of $A(n) = \prod_{p\in S} p^{e_p-1}$ is at least 1, and for $p \in T$ we have $p^{e_p-1} \geq p$. Hence $A(n) \geq \prod_{p\in T} p^{e_p-1} \geq \prod_{p\in T} p$. (iii) follows from (ii). □

We write $P(q) = \prod_{p<q} p$, and $\mathcal{P}(x)$ for the set of primes $p < x$.

3. Reduction to finitely many shapes for each q

Lemma 3.1. *Let $n \geq 1$ with $A(n) < q(n)$, and put $q = q(n)$ and $a = A(n)$. Then $1 \leq a < q$, $v_p(n) = 1 + v_p(a)$ for every prime $p < q$, and*

$$2\,a\,P(q) \;\leq\; G(q,a) := \prod_{p<q}(1+p^{1+v_p(a)}) + \prod_{p<q}(1+p) \tag{2}$$

implies $\sigma^*(n) + \sigma_\square(n) \geq 2n$.

Proof. By Lemma 2.3, $S = \mathcal{P}(q) \sqcup S_1$, where S_1 is the set of prime divisors of n greater than q, and $e_p = 1$ for $p \in S_1$. Hence $a = A(n) = \prod_{p<q} p^{e_p - 1}$, and unique factorization gives $v_p(a) = e_p - 1$ for $p < q$. Write $n = n_0\, s$ with $n_0 = \prod_{p<q} p^{e_p} = a\, P(q)$ and $s = \prod_{p \in S_1} p$. Using $e_p = 1$ on S_1, Lemma 2.1 gives

$$\sigma^*(n) + \sigma_\square(n) = \Big(\prod_{p<q}(1 + p^{e_p}) + \prod_{p<q}(1+p)\Big) \prod_{p \in S_1}(1+p) = G(q,a) \prod_{p \in S_1}(1+p).$$

Since $\prod_{p \in S_1}(1+p) \geq s$, inequality (2) gives $\sigma^*(n) + \sigma_\square(n) \geq 2aP(q)s = 2n$. □

Remark 3.2. Conversely, for every prime q and every $1 \leq a < q$, the number $n_0 = aP(q)$ satisfies $A(n_0) = a$ and $q(n_0) = q$, since all prime factors of a are smaller than q. Hence (2) for all such pairs (q, a) is *equivalent* to Theorem 1.1.

4. The case $q(n) \leq 73$

Proposition 4.1. *Inequality (2) holds for every prime $q \leq 73$ and every integer $1 \leq a < q$. Equality holds exactly for $(q,a) \in \{(2,1),(3,2),(5,4)\}$.*

Proof. This is a finite verification over $\sum_{q \leq 73}(q-1) = 691$ pairs (q,a), where the sum runs over the 21 primes $q \leq 73$. Each case is an inequality between explicit integers. In the formal proof (Section 7), all 691 cases are checked by the Lean kernel itself (tactic `decide`), with $v_p(a)$ computed by a recursive function proved equal to the p-adic valuation. The equality cases were determined by an independent exact computation in rational arithmetic (script `checks/equality_cases.py`, which also reproduces Tables 1 and 2). Table 1 lists, for each q, the minimum of $G(q,a)/(aP(q))$ over $1 \leq a < q$. □

q	min	at a	q	min	at a	q	min	at a
2	2.0000	1	19	2.4010	18	47	2.6328	36
3	2.0000	2	23	2.5141	20	53	2.6125	48
5	2.0000	4	29	2.3683	24	59	2.6618	48
7	2.0667	6	31	2.4500	24	61	2.5454	60
11	2.2552	10	37	2.4456	36	67	2.5871	60
13	2.1195	12	41	2.5117	36	71	2.6257	60
17	2.2825	12	43	2.5730	36	73	2.6627	60

Table 1. $\min_{1 \leq a < q} G(q,a)/(aP(q))$ for the primes $q \leq 73$, rounded to four decimals.

5. The case $q(n) > 73$

In this case we show that $\sigma^*(n) \geq 2n$ on its own. Let $p_1 = 2 < p_2 = 3 < \cdots$ be the primes in increasing order. For $t \geq 0$ put

$$R_t = \{p_1, \ldots, p_t\}, \qquad P_t = \prod_{i=1}^{t} p_i, \qquad D_t = \prod_{i=1}^{t}\Big(1 + \frac{1}{p_i}\Big),$$

and for an integer $x \geq 1$ put

$$N(x) = \prod_{p<x}\Big(1 + \frac{1}{p}\Big), \qquad H_m = \sum_{k=1}^{m} \frac{1}{k} \quad (H_0 = 0).$$

Lemma 5.1. *Let $n \geq 1$ with $A(n) < q(n) = q$, and let T be as in Lemma 2.3. Then*

$$\frac{\sigma^*(n)}{n} \;\geq\; \prod_{\substack{p<q \\ p \notin T}}\Big(1 + \frac{1}{p}\Big).$$

Proof. By Lemma 2.1, $\sigma^*(n)/n = \prod_{p\in S}(1+p^{-e_p})$. Every factor is at least 1, so the product is at least $\prod_{p\in S\setminus T}(1+p^{-e_p}) = \prod_{p\in S\setminus T}(1+1/p)$. By Lemma 2.3(i), $\mathcal{P}(q)\setminus T \subseteq S\setminus T$, and the claim follows, again because all factors are at least 1. □

Lemma 5.2 (Rearrangement)**.** *Let T be a finite set of primes and $t = |T|$. Then*

$$(a)\quad \prod_{p\in T}\Big(1+\frac{1}{p}\Big) \le D_t, \qquad\qquad (b)\quad P_t \le \prod_{p\in T} p.$$

Proof. Since $|T| = |R_t| = t$, we have $|T\setminus R_t| = |R_t\setminus T| =: c$. Every element of $R_t\setminus T$ is at most p_t. Every element of $T\setminus R_t$ is a prime outside $\{p_1,\dots,p_t\}$, hence greater than p_t. Therefore

$$\prod_{p\in T\setminus R_t}\Big(1+\frac{1}{p}\Big) \le \Big(1+\frac{1}{p_t}\Big)^c \le \prod_{p\in R_t\setminus T}\Big(1+\frac{1}{p}\Big), \qquad \prod_{p\in R_t\setminus T} p \le p_t^c \le \prod_{p\in T\setminus R_t} p.$$

Multiplying both inequalities by the common factor over $T\cap R_t$ gives (a) and (b). (For $t=0$ both sides equal 1.) □

Lemma 5.3. *For every integer $x\ge 1$, $N(x) \ge \frac{1}{2}H_{x-1}$.*

Proof. Let $y = x-1$, and let $Q(y)$ be the set of squarefree integers d with $1\le d\le y$. Expanding the product,

$$N(x) = \sum_{U\subseteq\mathcal{P}(x)}\ \prod_{p\in U}\frac{1}{p}.$$

The map sending $d\in Q(y)$ to its set of prime factors is injective, since a squarefree d is the product of its prime factors. Its values are subsets of $\mathcal{P}(x)$, since each prime factor of d is at most $d\le y < x$. Moreover $\prod_{p\mid d} 1/p = 1/d$. All terms of the sum being nonnegative, we get $N(x)\ge \sum_{d\in Q(y)} 1/d$.

Every integer $1\le m\le y$ can be written as $m = d\,j^2$ with d squarefree and $j\ge 1$. Choosing one such pair (d_m, j_m) for each m defines an injective map $m\mapsto(d_m,j_m)$, since $m = d_m j_m^2$ is recovered from it, with values in $Q(y)\times\{1,\dots,y\}$. Hence, again by nonnegativity,

$$H_y = \sum_{m=1}^{y}\frac{1}{m} = \sum_{m=1}^{y}\frac{1}{d_m\,j_m^2} \le \Big(\sum_{d\in Q(y)}\frac{1}{d}\Big)\Big(\sum_{j=1}^{y}\frac{1}{j^2}\Big) \le 2\sum_{d\in Q(y)}\frac{1}{d},$$

because $\sum_{j=1}^{y} j^{-2} \le 1 + \sum_{j=2}^{y}\frac{1}{j(j-1)} = 2-\frac{1}{y} < 2$ (for $y\ge 1$; for $y=0$ both sides vanish). Combining the two estimates gives $N(x)\ge\frac{1}{2}H_{x-1}$. □

Lemma 5.4. *For every integer $N\ge 1$, $H_{2N}\ge H_N+\frac{1}{2}$. Consequently $H_{2^m}\ge 1+\frac{m}{2}$ for all $m\ge 0$, and $H_{16N}\ge H_N+2$ for all $N\ge 1$. Moreover, $m\le m'$ implies $H_m\le H_{m'}$.*

Proof. $H_{2N}-H_N = \sum_{k=N+1}^{2N}\frac{1}{k} \ge N\cdot\frac{1}{2N} = \frac{1}{2}$. The other claims follow by iterating this (from $H_1=1$), and monotonicity is clear. □

Lemma 5.5. *For every $t\ge 0$, $D_t\le (t+2)/2$.*

Proof. By induction on t. The case $t=0$ is $D_0=1$. Since the i-th prime satisfies $p_i\ge i+1$, we have $p_{t+1}\ge t+2$, and therefore $D_{t+1} = D_t\,(1+1/p_{t+1}) \le \frac{t+2}{2}\cdot\frac{t+3}{t+2} = \frac{t+3}{2}$. □

Lemma 5.6. *For every $t\ge 10$, $H_{P_t}\ge 4D_t$.*

Proof. By induction on $t\ge 10$. For $t=10$: $P_{10} = 6\,469\,693\,230 \ge 2^{32}$, so by Lemma 5.4 $H_{P_{10}}\ge H_{2^{32}}\ge 17$, while

$$D_{10} = \prod_{p\le 29}\Big(1+\frac{1}{p}\Big) = \frac{119\,439\,360}{30\,808\,063} < 3.877, \qquad 4D_{10} < 15.51 < 17.$$

For the induction step let $t \geq 10$ and $p = p_{t+1}$. Then $p \geq p_{11} = 31 \geq 16$, so $P_{t+1} = p\,P_t \geq 16P_t$, and Lemma 5.4 gives $H_{P_{t+1}} \geq H_{16P_t} \geq H_{P_t} + 2$. On the other hand, by Lemma 5.5 and $p \geq t+2$, we have $D_t \leq (t+2)/2 \leq p/2$, hence

$$4D_{t+1} = 4D_t + \frac{4D_t}{p} \leq 4D_t + 2 \leq H_{P_t} + 2 \leq H_{P_{t+1}}. \qquad \square$$

Proposition 5.7. *Let $q > 73$ and $t \geq 0$ be integers with $P_t < q$. Then $N(q) \geq 2D_t$.*

Proof. Case $t \geq 10$. By Lemmas 5.3, 5.4 and 5.6, since $q - 1 \geq P_t$, $N(q) \geq \frac{1}{2}H_{q-1} \geq \frac{1}{2}H_{P_t} \geq 2D_t$.

Case $t \leq 9$. $N(x)$ is nondecreasing in x. For each $t \leq 9$ we choose a threshold y_t with $y_t \leq q$: for $t \leq 3$ we use $y_t = 74 \leq q$; for $4 \leq t \leq 9$ the value y_t given in Table 2 satisfies $y_t \leq P_t + 1 \leq q$. In every case $N(q) \geq N(y_t) \geq 2D_t$, where the last inequality is an explicit comparison of two finite products of rational numbers, recorded in Table 2 and checked exactly in the formal proof. $\square$

t	p_t	P_t	y_t	$N(y_t)$	$2D_t$
0	–	1	74	4.83440	2.00000
1	2	2	74	4.83440	3.00000
2	3	6	74	4.83440	4.00000
3	5	30	74	4.83440	4.80000
4	7	210	150	5.50182	5.48571
5	11	2 310	234	5.98744	5.98442
6	13	30 030	368	6.44929	6.44476
7	17	510 510	522	6.82594	6.82386
8	19	9 699 690	734	7.18488	7.18301
9	23	223 092 870	992	7.50192	7.49531

TABLE 2. Thresholds used for $t \leq 9$ in Proposition 5.7 (values rounded to five decimals; the comparisons $N(y_t) \geq 2D_t$ are verified in exact arithmetic).

Theorem 5.8. *Let $n \geq 1$ with $A(n) < q(n)$ and $q(n) > 73$. Then $\sigma^*(n) \geq 2n$.*

Proof. Let $q = q(n)$, let T be as in Lemma 2.3, and let $t = |T|$. By Lemma 2.3(ii), $T \subseteq \mathcal{P}(q)$ and $\prod_{p\in T} p < q$, so Lemma 5.2(b) gives $P_t < q$. Proposition 5.7 then gives $N(q) \geq 2D_t$, and Lemma 5.2(a) gives

$$\prod_{\substack{p<q \\ p\notin T}} \Big(1 + \frac{1}{p}\Big) = \frac{N(q)}{\prod_{p\in T}(1+1/p)} \geq \frac{N(q)}{D_t} \geq 2.$$

By Lemma 5.1, $\sigma^*(n) \geq 2n$. $\square$

6. Proof of the main theorem and the equality cases

Proof of Theorem 1.1. Let $n \geq 1$ with $A(n) < q(n)$. If $q(n) > 73$, Theorem 5.8 gives $\sigma^*(n) + \sigma_\square(n) > \sigma^*(n) \geq 2n$. If $q(n) \leq 73$, Proposition 4.1 shows that (2) holds for the pair $(q(n), A(n))$, and Lemma 3.1 gives $\sigma^*(n) + \sigma_\square(n) \geq 2n$. $\square$

Corollary 6.1. *Let $n \geq 1$ with $A(n) < q(n)$. Then $\sigma^*(n) + \sigma_\square(n) = 2n$ if and only if $n \in \{1, 4, 24\}$.*

Proof. If $q(n) > 73$, the inequality is strict (see the proof above). If $q(n) \leq 73$, the proof of Lemma 3.1 gives

$$\sigma^*(n) + \sigma_\square(n) = G(q,a) \prod_{p\in S_1}(1+p) \geq 2aP(q) \prod_{p\in S_1}(1+p) \geq 2aP(q)\,s = 2n.$$

Equality forces $S_1 = \varnothing$ (so $s = 1$) and $G(q, a) = 2aP(q)$. By Proposition 4.1 this means $(q, a) \in \{(2, 1), (3, 2), (5, 4)\}$, that is, $n = aP(q) \in \{1, 4, 24\}$. Conversely, $\sigma^*(1) + \sigma_\square(1) = 2$, $\sigma^*(4) + \sigma_\square(4) = 8$, and $\sigma^*(24) + \sigma_\square(24) = 36 + 12 = 48$. □

Remark 6.2. Corollary 6.1 uses the equalities of Lemma 2.1 and the equality cases of Proposition 4.1. It was not part of the formalization, which covers Theorem 1.1 only.

7. Formal verification

Theorem 1.1 (but not Corollary 6.1) has been formalized in Lean 4 (version `4.36.0-rc2`) with the Mathlib library (commit `9d581dfff2e41de989072ea63abf3daf69311520`), in a single file `A401714.lean` of about 870 lines, which can be checked inside any Lean project depending on that version of Mathlib by running

`lake env lean A401714.lean`.

The final statement is:

```
def usigma (n : ℕ) : ℕ := ∑ d ∈ n.divisors.filter (fun d => Nat.Coprime d (n / d)), d
def ssq (n : ℕ) : ℕ := ∑ d ∈ n.divisors.filter (fun d => Squarefree d), d
def rad (n : ℕ) : ℕ := ∏ p ∈ n.primeFactors, p
noncomputable def A003557 (n : ℕ) : ℕ := n / rad n
noncomputable def A053669 (n : ℕ) : ℕ :=
  if hn : 0 < n then Nat.find (exists_prime_not_dvd n hn) else 2

theorem A401714_le_A401715 (n : ℕ) (hn : 0 < n) (h : A003557 n < A053669 n) :
    2 * n ≤ usigma n + ssq n
```

Here `Nat.find` returns the least natural number with the stated property, so `A053669 n` is the smallest prime not dividing n; a lemma in the accompanying checks proves this characterization explicitly. The following points were verified.

(1) **No gaps.** The file compiles without errors and contains no `sorry`. The command `#print axioms A401714.A401714_le_A401715` reports only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. In particular `native_decide` is not used: the finite computations of Proposition 4.1 and Table 2 are checked by the Lean kernel through `decide` and `norm_num`.

(2) **Fidelity of the statement.** The Lean definitions were evaluated and compared with the OEIS data: `usigma` reproduces A034448 (70 terms), `ssq` reproduces A048250 (73 terms), `n / rad n` reproduces A003557 (98 terms), the smallest prime not dividing n reproduces A053669 (105 terms), and the indicator functions of the hypothesis and of the conclusion reproduce A401714 and A401715 (105 terms each).

(3) **Non-vacuity.** The hypotheses of the theorem are shown, inside Lean, to be satisfiable (for example $n = 24$, $q = 5$).

(4) **Independent re-check.** The file was compiled a second time, independently, from an unmodified copy with SHA-256 checksum

`1a5db0e926d289257a65a430d5b7d4f18e8f2f8b922f484d18ca46b3c3200d03`.

The published file `A401714.lean` differs from that copy only in an attribution comment on line 6 (no Lean code is affected); its SHA-256 checksum is

`057a6b428221249d2e7b49048b477d7d2fe1720c5a92c47947cc69a69e3b2649`.

The formal structure follows the paper: the product lower bounds of Lemma 2.1; the reduction of Lemma 3.1; the kernel-checked finite verification for $q \le 73$; Lemmas 5.2–5.6; and Proposition 5.7 and Theorem 5.8.

Acknowledgements and disclosure

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the rearrangement bound of Lemma 5.2, a log-free form of the induction in Lemma 5.6, exact recomputation of the numerical margins, and an independent audit of the final formal proof. The author, writing under the pseudonym A-guy-with-AI, is responsible for the statement and its presentation. The conjecture is due to Antti Karttunen. The Lean source files, the paper's source, and the computation scripts are available at https://github.com/A-guy-with-AI/A401714-proof.

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